Image tensors and channels

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Image tensors and channels before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Learned representations

Without opening the old lesson, explain one key idea from Learned representations. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Initialisation and reproducibility

Without opening the old lesson, explain one key idea from Initialisation and reproducibility. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Learning rate experiments

Without opening the old lesson, explain one key idea from Learning rate experiments. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

A small fictional RGB image has height 2 pixels, width 2 pixels and three **channel**s ordered red, green, blue. Store it as image[row][column][**channel**], abbreviated HWC.

Use zero-based indices: the top row is 0, the left column is 0, and red, green and blue are **channel**s 0, 1 and 2. Each stored component is an integer from 0 to 255.

Position	RGB triplet	Meaning
Row 0 column 0	[255, 0, 0]	Red
Row 0 column 1	[0, 255, 0]	Green
Row 1 column 0	[0, 0, 255]	Blue

Position	RGB triplet	Meaning
Row 1 column 1	[255, 255, 255]	White

The shape is 2 by 2 by 3. There are four pixels and twelve **channel** values. The blue component of the bottom-left pixel is image[1][0][2] = 255.

A single **channel** is a plane of values across positions; it is not a separate photograph or a label.

Shape needs a schema. A grayscale image might be stored as H by W or H by W by 1. An RGB image uses three **channel**s here; other formats can include additional **channel**s.

Declare the format instead of assuming every three-dimensional array means the same thing.

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Image tensors and channels trace and record the first step you can explain without help.

Use W2: Audit an axis conversion for Image **tensor**s and **channel**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Image tensors and channels explanation against the supplied facts before discussing it.

Use the worked case for Image **tensor**s and **channel**s. Proposed explanation: “Three **channel**s mean three images”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Convert a new image independently

A one-row two-column RGB image has pixels [10, 20, 30] then [40, 50, 60]. State its HWC shape and all three CHW planes. Give the green value at row 0 column 1 in both layouts.

A batch contains four copies: give its NHWC shape and total **channel**-value count.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Read a pixel precisely

> Read a pixel precisely. Apply the supplied Image tensors and channels facts and keep your reasoning.

For the lesson image, give image[1][0], image[1][0][2], the HWC shape, pixel count and number of **channel** values.

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Audit an axis conversion

> Audit an axis conversion. Apply the supplied Image tensors and channels facts and keep your reasoning.

A program flattens the lesson HWC array and groups the first four values into a 2 by 2 red plane. Give that incorrect plane and the correct plane. Identify a position where they disagree.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Check scaling and lost colour

Check scaling and lost colour. Apply the supplied Image tensors and channels facts and keep your reasoning.

Scale [255, 128, 0] by 255, keeping the middle value as a fraction. Then calculate the classroom RGB means of pure red and pure green. Which operation loses the distinction between those colours?

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Convert a new image independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Convert a new image independently. Apply the supplied Image tensors and channels facts and keep your reasoning.

A one-row two-column RGB image has pixels [10, 20, 30] then [40, 50, 60]. State its HWC shape and all three CHW planes. Give the green value at row 0 column 1 in both layouts.

A batch contains four copies: give its NHWC shape and total **channel**-value count.

> **My reasoning and evidence**

W5 Design revealing pipeline checks

> Design revealing pipeline checks. Apply the supplied Image tensors and channels facts and keep your reasoning.

The lesson image is loaded with red and blue exchanged, then scaled twice. Give two expected checks that would expose these errors even though the shape is correct.

> **My reasoning and evidence**

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Image tensors and channels to Image-pattern Network Investigation, identify a limit and choose the next useful test.

Add a labelled image **tensor**s and **channel**s **evidence** sheet to Image-pattern Network Investigation. Show how you can read a pixel using declared row column and **channel** axes.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	

Evidence	My record
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

Pictorial case practice: Image tensors and channels

> Move colour values between tensor layouts without changing the pixels. Explain the labelled step before checking the changed condition.

Move colour values between **tensor** layouts without changing the pixels.

Toy RGB image has 2 rows, 1 column. Top pixel [1, 4, 7]; bottom [2, 5, 8]. HWC means height, width, **channel**.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/20

Two RGB pixels before layout conversion

	red	green	blue
top pixel	1	4	7
bottom pixel	2	5	8

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/20

Image tensors and channels

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Record HWC shape

Shape is [2, 1, 3]; top green is 4 and bottom green 5.

2. Reorder to CHW

Shape [3, 2, 1]; red plane [[1],[2]], green [[4],[5]], blue [[7],[8]].

3. Check same pixel

Bottom green is HWC [1, 0, 1]=5 and CHW [1, 1, 0]=5.

CHANGE ONE THING

Add a batch containing three identical images in NHWC layout.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Where is the bottom green pixel in the first batched image? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

WORKBOOK · 13 OF 15

Project portfolio milestone

> Keep this week’s input, deciding step, result and limit together in the Image-pattern Network Investigation record.

Project: Image-pattern Network Investigation

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Move colour values between **tensor** layouts without changing the pixels.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Bottom green is HWC [1, 0, 1]=5 and CHW [1, 1, 0]=5.

1. Record HWC shape

2. Reorder to CHW

3. Check same pixel

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Check same pixel Bottom green is HWC [1, 0, 1]=5 and CHW [1, 1, 0]=5. A **tensor** is an array with named dimensions.

HWC orders height, width and colour; CHW puts colour first. Rearranging the layout changes where values are stored, not the colour measurement itself. Trace a named pixel through both layouts using zero-based indices.

Adding a batch dimension counts images and does not add a new colour **channel**.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Add a batch containing three identical images in NHWC layout. Which conclusion is supported by the supplied facts?

1. Batch shape is [3, 2, 1, 3].

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Batch shape is [3, 2, 1, 3]. The new first dimension counts images; it does not add a fourth colour **channel**.

Repair a missing building block

Review Learned representations (ai-5/19). Goal: Trace an **input** through a hidden representation and a decision rule. Short practice: Why can no deterministic head using only the sum classify both correctly?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Initialisation and reproducibility (ai-5/17). Goal: Trace how starting values affect a fixed **training** budget. Short practice: Does the matching final loss mean the parameter values are equal?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Learning rate experiments (ai-5/14). Goal: Run a controlled comparison of learning rates. Short practice: What should a fair rate comparison hold fixed?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

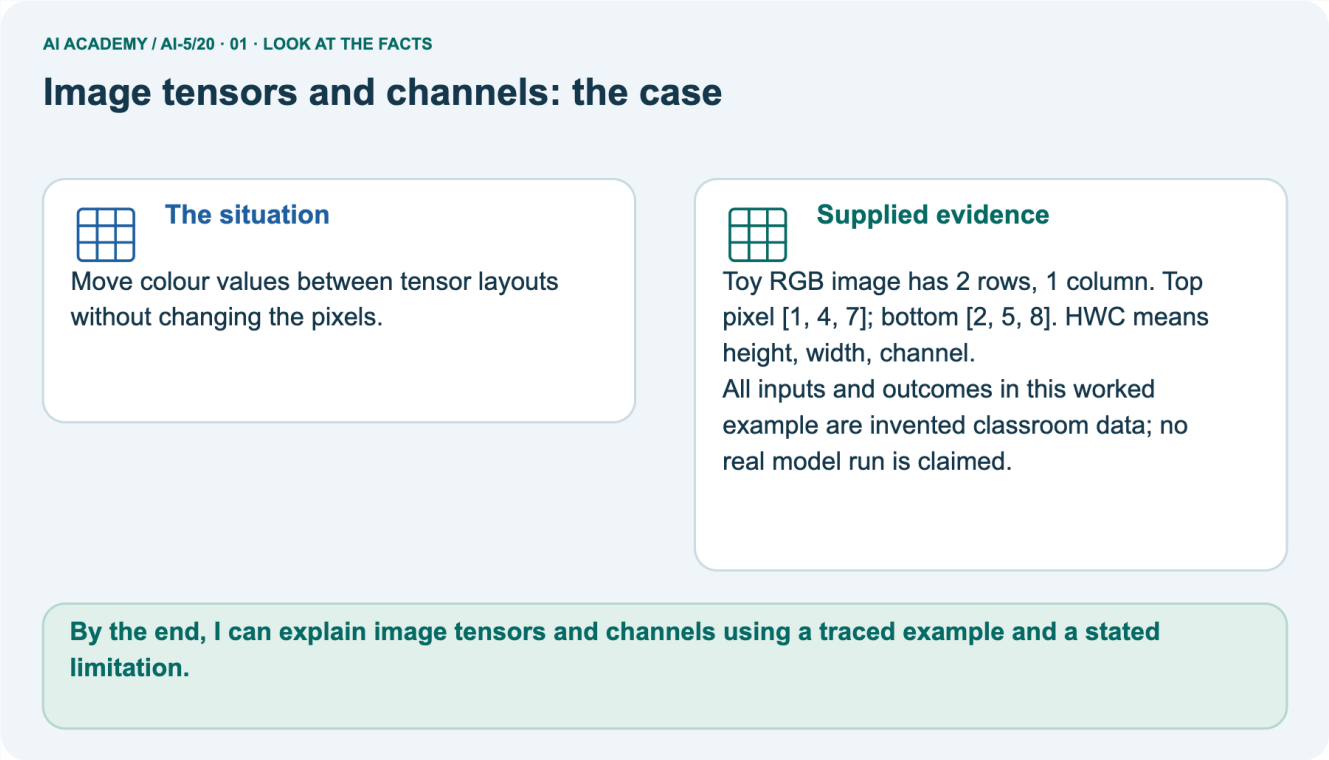
Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**

Visual concept journey

> Explain each Image tensors and channels picture from its labels, then record a relevant limit.

01. Image tensors and channels: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

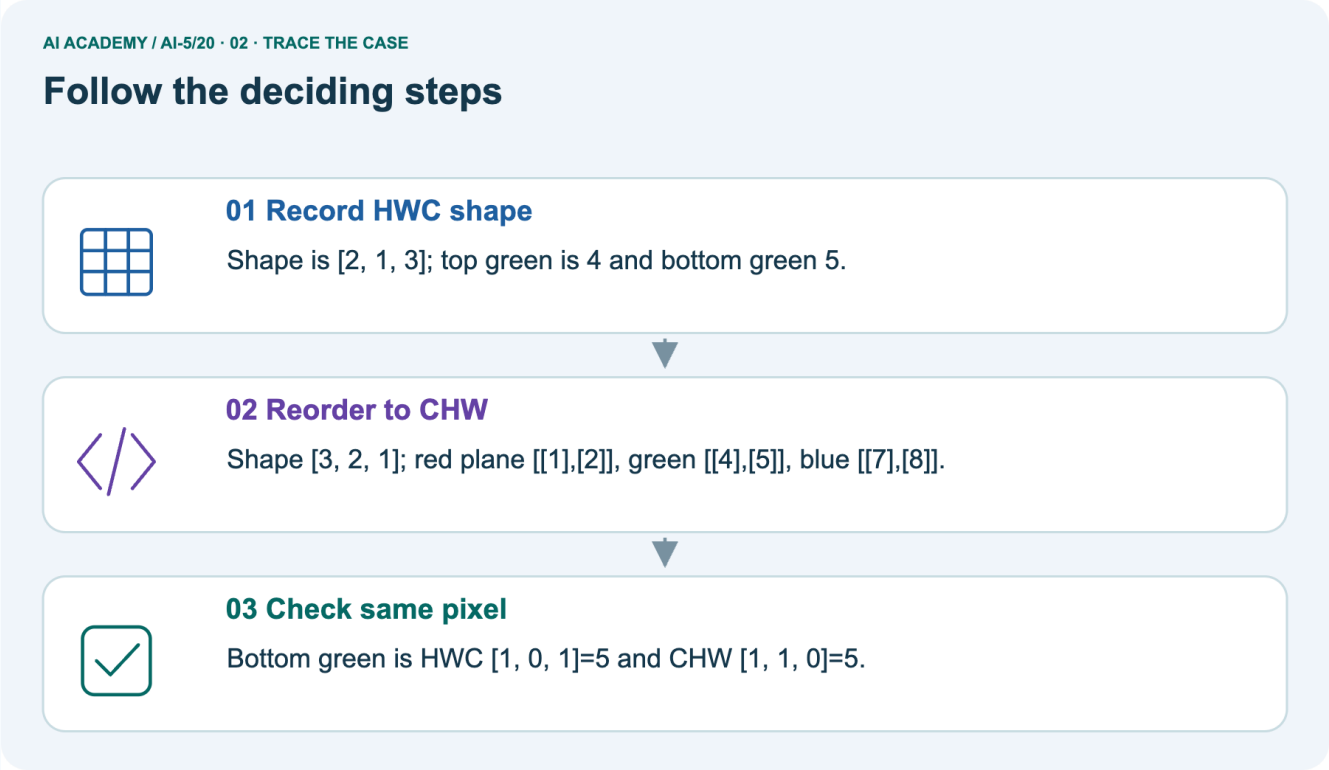
Move colour values between **tensor** layouts without changing the pixels. Supplied facts: Toy RGB image has 2 rows, 1 column. Top pixel [1, 4, 7]; bottom [2, 5, 8]. HWC means height, width, **channel**.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



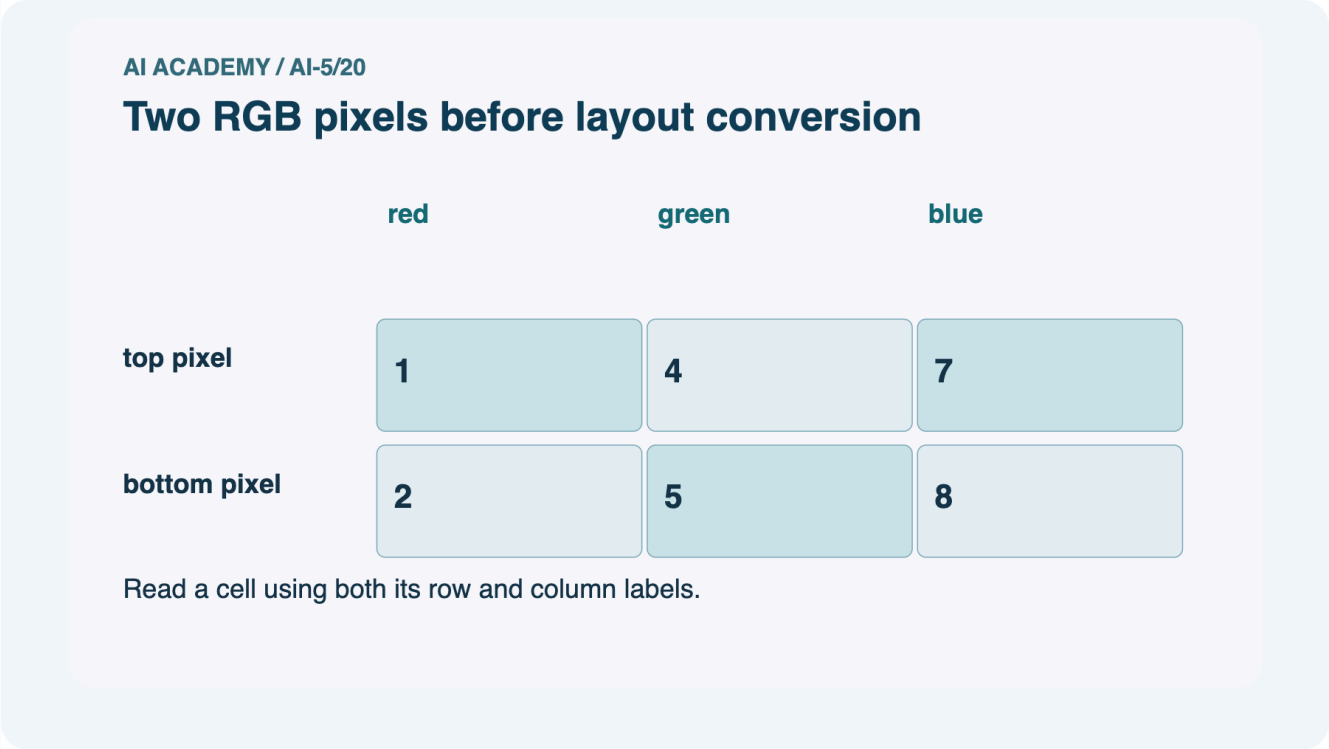
Follow the labelled picture with the matching explanation. Open the vector diagram

Record HWC shape: Shape is [2, 1, 3]; top green is 4 and bottom green 5. → Reorder to CHW: Shape [3, 2, 1]; red plane [[1],[2]], green [[4],[5]], blue [[7],[8]]. → Check same pixel: Bottom green is HWC [1, 0, 1]=5 and CHW [1, 1, 0]=5.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Two RGB pixels before layout conversion



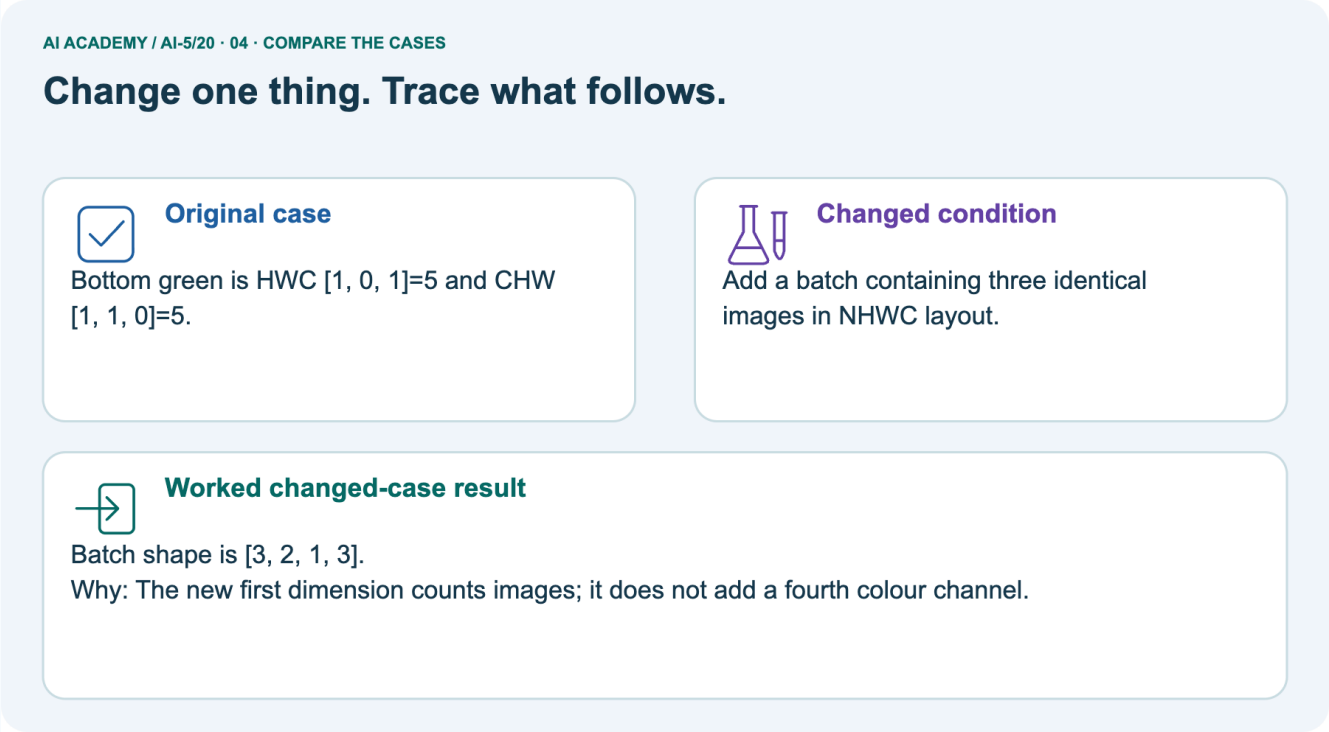
Follow the labelled picture with the matching explanation. Open the vector diagram

Two RGB pixels before layout conversion. top pixel: red = 1, green = 4, blue = 7. bottom pixel: red = 2, green = 5, blue = 8.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

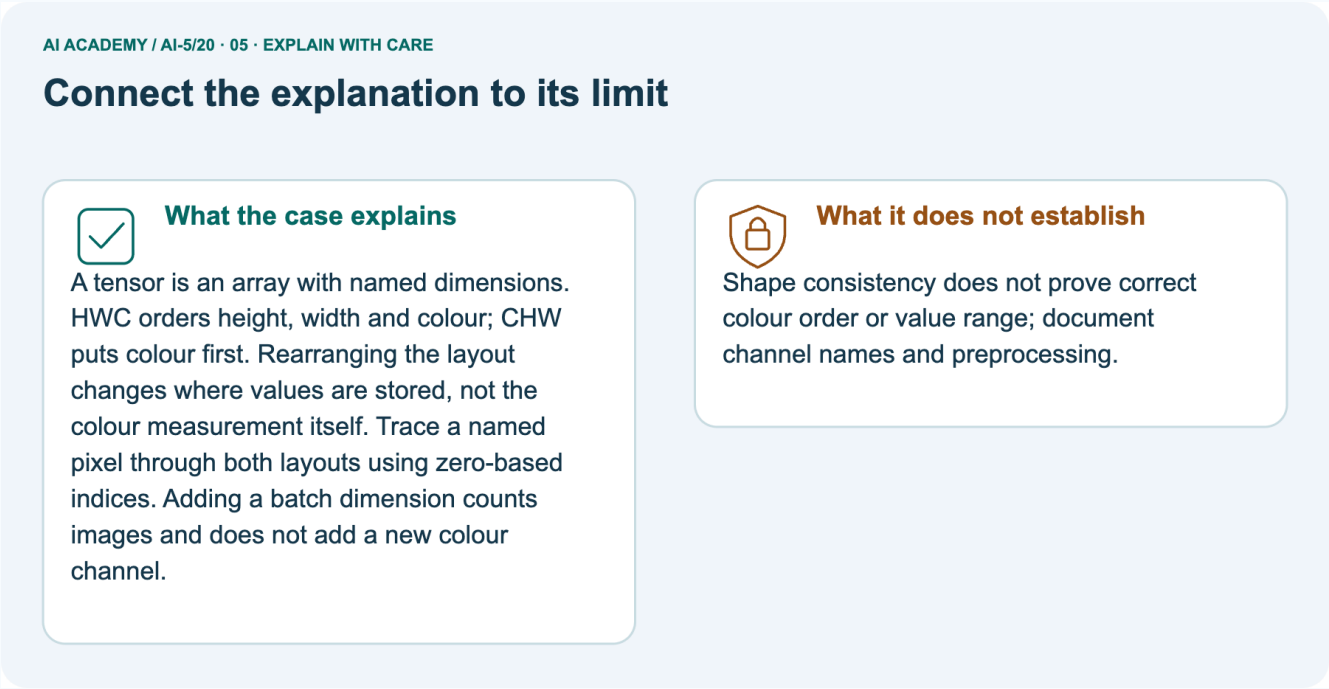
Original case: Bottom green is HWC [1, 0, 1]=5 and CHW [1, 1, 0]=5. Changed condition: Add a batch containing three identical images in NHWC layout. Expected result: Batch shape is [3, 2, 1, 3].

Reason: The new first dimension counts images; it does not add a fourth colour **channel**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **tensor** is an array with named dimensions. HWC orders height, width and colour; CHW puts colour first. Rearranging the layout changes where values are stored, not the colour measurement itself.

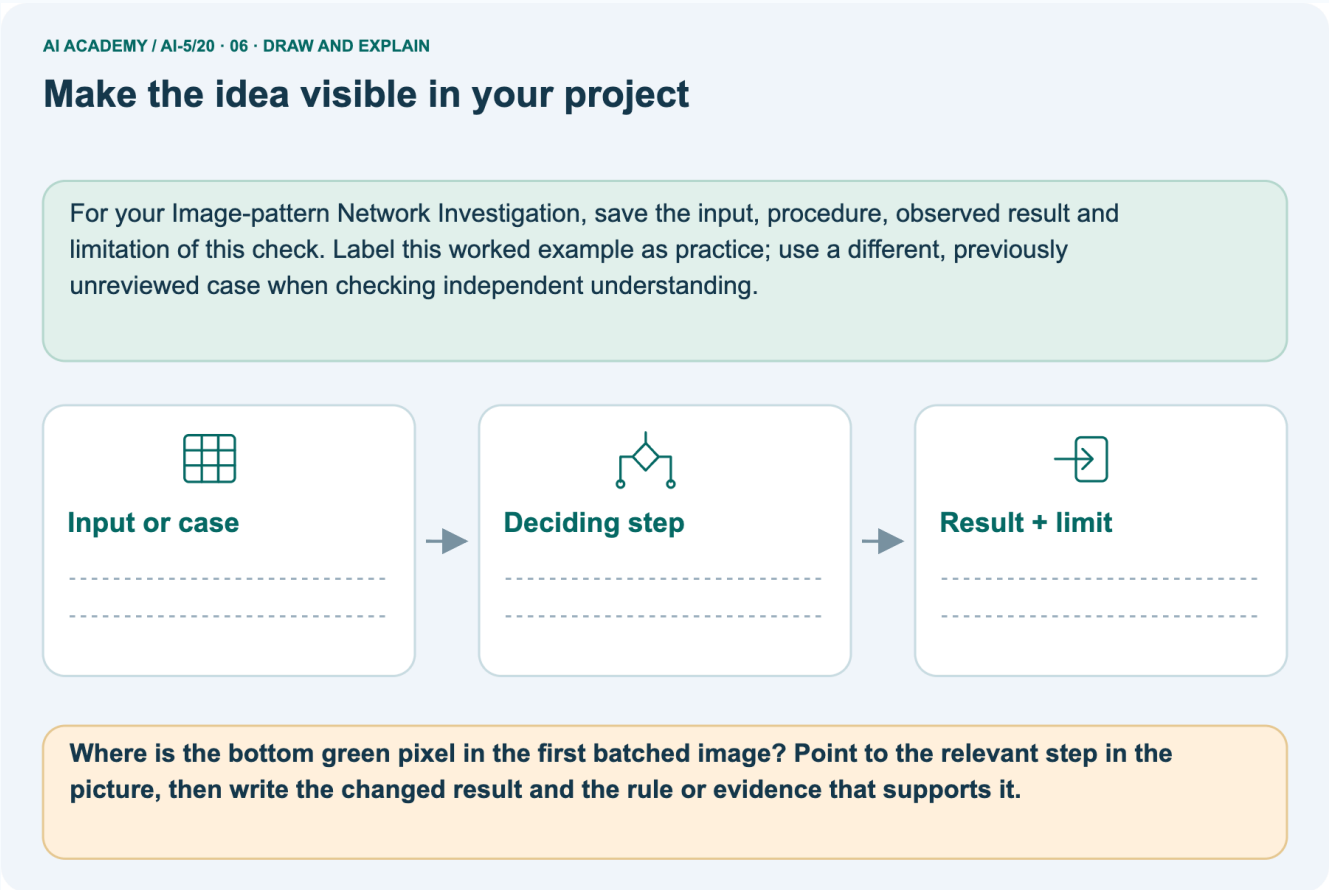
Trace a named pixel through both layouts using zero-based indices. Adding a batch dimension counts images and does not add a new colour **channel**.

Limit: Shape consistency does not prove correct colour order or value range; document **channel** names and pre**process**ing.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Where is the bottom green pixel in the first batched image? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.